Cells, Genomes, and the Diversity of Life

The Universal Features of Life on Earth

All living organisms are fundamentally similar. They are made of cells, small membrane-enclosed units filled with a concentrated aqueous solution of chemicals, with the ability to create copies of themselves.

All Cells Store Their Hereditary Information in DNA

- **Heredity:** The central feature of life, where an organism passes down information specifying its characteristics to its offspring.
- **Universal Storage Medium:** All cells on Earth store their hereditary information in double-strand DNA molecules. This suggests a common ancestor for all life, which existed approximately 3.5-3.8 billion years ago.
- **Genome:** The totality of an organism’s hereditary information, encoded in the linear sequence of nucleotides (A, T, C, G) in its DNA.
- **Interoperability:** DNA from one species (e.g., human) can be inserted into another (e.g., bacterium) and be read, interpreted, and copied.
- **Figure 1-1:** Illustrates how the hereditary information in a fertilized egg determines the nature of the resulting multicellular organism (e.g., sea urchin, mouse, seaweed).

All Cells Replicate Their Hereditary Information by Templated Polymerization

- **DNA Structure:** A DNA strand is a polymer of nucleotides. Each nucleotide consists of a sugar-phosphate backbone and a base (A, G, C, or T).
- **Base Pairing:** The two strands of a DNA double helix are held together by hydrogen bonds between complementary bases: Adenine (A) pairs with Thymine (T), and Guanine (G) pairs with Cytosine (C).
- **Templated Polymerization:** The process of DNA replication. The two strands of the DNA double helix are separated, and each strand serves as a template for the synthesis of a new, complementary strand.
- **Mechanism:** The sequence of the template strand dictates the sequence of the new strand according to the base-pairing rules. This ensures the faithful copying of genetic information.
- **Figure 1-2:** Shows the structure of DNA, its nucleotide building blocks, and the process of templated polymerization.
- **Figure 1-3:** Depicts how DNA replication results in two identical daughter DNA double helices from one parent molecule.

All Cells Transcribe Portions of Their DNA into RNA

- **Gene Expression:** The process by which the information stored in DNA is used to synthesize other molecules.
- **Transcription:** The first step of gene expression, where segments of the DNA sequence are used as templates to synthesize RNA (ribonucleic acid) molecules.
- **RNA:** A polymer similar to DNA, but with ribose sugar instead of deoxyribose and Uracil (U) instead of Thymine (T). U pairs with A.
- **Messenger RNA (mRNA):** Most RNA transcripts serve as mRNA, which guides the synthesis of proteins.

- **Other RNA Functions:** Some RNA molecules have catalytic, regulatory, or structural functions directly.
- **Figure 1-4:** Illustrates the flow of genetic information from DNA to RNA (transcription) and then to protein (translation).

All Cells Use Proteins as Catalysts

- **Proteins:** Long, unbranched polymers of amino acids. They are the main molecules that put the cell's genetic information into action.
- **Amino Acids:** The building blocks of proteins. There are 20 common types, each with a unique side group that gives it a specific chemical character.
- **Function from Structure:** The amino acid sequence of a protein (polypeptide) determines its unique three-dimensional folded structure, which in turn determines its function.
- **Enzymes:** Many proteins act as enzymes, which are catalysts that speed up specific chemical reactions without being consumed.
- **Autocatalysis:** Life is an autocatalytic process. DNA and RNA provide the information to make proteins, and proteins provide the catalytic activity to synthesize DNA, RNA, and other proteins.
- **Figure 1-5:** Shows life as a self-replicating collection of catalysts, highlighting the feedback loops between polynucleotides and proteins.

All Cells Translate RNA into Protein in the Same Way

- **Translation:** The process of converting the information from the 4-letter alphabet of RNA into the 20-letter alphabet of proteins.
- **Genetic Code:** The rules of translation. The sequence of an mRNA molecule is read in groups of three nucleotides, called codons.
- **Codons:** Each codon specifies one amino acid. There are 64 possible codons (4^3), but only 20 amino acids, so the code is redundant (multiple codons for one amino acid).
- **Transfer RNA (tRNA):** A class of small RNA molecules that act as adaptors. Each tRNA has an anticodon that recognizes a specific mRNA codon and carries the corresponding amino acid.
- **Ribosome:** A large macromolecular machine, composed of ribosomal RNA (rRNA) and proteins, that carries out protein synthesis.

Each Protein Is Encoded by a Specific Gene

- **Gene:** A segment of DNA sequence that codes for a single protein (or a set of related protein variants) or a single functional RNA molecule (like tRNA or rRNA).
- **Gene Regulation:** Cells control the expression of genes, adjusting the rate of transcription and translation according to their needs. Regulatory DNA sequences, interspersed with coding segments, bind to transcription regulator proteins to control gene activity.

Life Requires a Continual Input of Free Energy

- **Non-Equilibrium System:** A living cell is a dynamic system operating far from chemical equilibrium.
- **Energy for Order:** Cells require a constant input of free energy from the environment to create and maintain order, drive synthetic reactions, and resist the disordering effects of thermal motion.
- **Energy Sources:**
  - **Animals:** Obtain free energy from the chemical bonds in food molecules.

- ▸ **Plants:** Obtain free energy from sunlight.

All Cells Function as Biochemical Factories
- All cells use a similar collection of small organic molecules (sugars, nucleotides, amino acids).
- **ATP (Adenosine Triphosphate):** A universally required nucleotide that serves as both a building block for DNA/RNA and the primary carrier of chemical energy for cellular reactions.

All Cells Are Enclosed in a Plasma Membrane
- **Plasma Membrane:** A selective barrier that encloses every cell, enabling it to concentrate nutrients, retain synthesized products, and excrete waste.
- **Amphiphilic Molecules:** The membrane is composed of molecules (mainly phospholipids) that have a hydrophilic (water-loving) head and a hydrophobic (water-fearing) tail.
- **Lipid Bilayer:** In water, phospholipids spontaneously aggregate to form a lipid bilayer, which can close to form vesicles. This is a key example of self-assembly.
- **Membrane Transport Proteins:** Specialized proteins embedded in the membrane control the passage of specific molecules into and out of the cell.
- **Figure 1-6:** Illustrates the amphiphilic nature of phospholipids and their self-assembly into a bilayer vesicle in water.

Cells Operate at a Microscopic Scale Dominated by Random Thermal Motion
- **Cellular Motion:** The cell interior is in perpetual motion. This includes directed movements by motor proteins and random thermal (Brownian) motion.
- **Brownian Motion:** The random movement of molecules due to collisions with other molecules. It drives diffusion and influences the rates of biochemical reactions.
- **Brownian Ratchet:** A mechanism where random thermal motion is harnessed for directed movement. For example, actin polymerization at the leading edge of a migrating cell can "ratchet" the plasma membrane forward.
- **Figure 1-7:** Explains how a Brownian ratchet mechanism can drive membrane protrusion.

A Living Cell Can Exist with 500 Genes
- **Minimal Genome:** The bacterium **Mycoplasma genitalium** has one of the smallest known genomes, with 525 genes in about 580,000 nucleotide pairs. This demonstrates that the basic requirements for life, while complex, are not unimaginably so.
- **Figure 1-8:** Shows an electron micrograph of **Mycoplasma genitalium**.

Genome Diversification and the Tree of Life
Genome analysis allows for the classification of all life based on evolutionary relationships, revealing a "tree of life."

The Tree of Life Has Three Major Domains: Eukaryotes, Bacteria, and Archaea
- **Molecular Phylogeny:** By comparing the DNA sequences of highly conserved genes, like those for ribosomal RNA (rRNA), we can determine the evolutionary distance between organisms.

- **Three Domains:** This analysis reveals three major divisions of life:
  1. **Bacteria:** A vast and diverse group of single-celled organisms.
  2. **Archaea:** Single-celled organisms that are visually similar to bacteria but genetically closer to eukaryotes.
  3. **Eukaryotes:** Organisms whose cells have a nucleus (e.g., plants, animals, fungi, protists).
- **Figure 1-9:** Presents a global tree of life based on genome comparisons, showing the three domains.

Eukaryotes, Bacteria, and Archaea

- **Eukaryotes:** (Greek for "truly nucleated"). Cells contain a membrane-bound nucleus and other organelles. They are generally larger and have larger genomes than prokaryotes.
- **Bacteria:** The most evolutionarily diverse domain. Typically small, single-celled organisms with a cell wall and a single cytoplasmic compartment. They exhibit immense biochemical diversity, inhabiting nearly every ecological niche.
  - **Figure 1-10 & 1-11:** Show the typical shapes, sizes, and simple internal structure of bacteria.
  - **Figure 1-12 & 1-13:** Illustrate the diversity of bacteria, including photosynthetic and chemosynthetic types.
- **Archaea:** The most poorly understood domain. They resemble bacteria in appearance but are genetically closer to eukaryotes. Many are extremophiles (living in extreme environments), but they are also abundant in soil and oceans.

Organisms Occupy Most of Our Planet

- **Biomass Distribution:** The total biomass on Earth is estimated at 550 gigatons of carbon (Gt C).
  - **Plants:** 450 Gt C (mostly terrestrial).
  - **Bacteria:** 70 Gt C (mostly in soil and Earth's crust).
  - **Archaea:** 7 Gt C.
  - **Animals:** 2 Gt C (mostly marine).
- **Figure 1-14:** A graph showing the proportional distribution of living biomass on Earth.

Cells Can Be Powered by a Wide Variety of Free-Energy Sources

- **Organotrophic:** Organisms that get energy by feeding on other living things or their organic products (e.g., animals, fungi).
- **Phototrophic:** Organisms that harvest energy from sunlight (e.g., plants, algae, many bacteria).
- **Lithotrophic:** Organisms that capture energy from energy-rich inorganic chemical systems (e.g., microbes at hydrothermal vents). They are "rock-feeders."
- **Hydrothermal Vents:** Deep-sea ecosystems powered by geochemical energy from chemicals like H_2S, H_2, and Fe^{2+}, supporting a dense population of lithotrophic microbes and the animals that feed on them.
- **Figure 1-15 & 1-16:** Depict a hydrothermal vent ecosystem and the organisms living there, such as giant tube worms.

Some Cells Fix Nitrogen and Carbon Dioxide for Other Cells

- **Nutrient Fixation:** The conversion of unreactive atmospheric N_2 and CO_2 into biologically usable organic compounds. This process requires a large amount of free energy.

- **Carbon Fixation:** Performed by phototrophs (like plants) and some lithotrophs.
- **Nitrogen Fixation:** Performed only by certain bacteria and archaea.
- **Symbiosis:** Organisms with complementary needs often form close associations. For example, nitrogen-fixing bacteria live in the root nodules of pea plants.

Genomes Diversify Over Evolutionary Time

- **Evolutionary Mechanism:** Evolution occurs through a cycle of mutation and natural selection.
  - **Mutation:** Random errors in DNA replication and repair alter the nucleotide sequence.
  - **Natural Selection:** Mutations that are beneficial are perpetuated; mutations that are damaging are eliminated; neutral mutations may or may not persist by chance.
- **Conserved Genes:** Genes coding for essential, highly optimized molecules (like rRNA) change very slowly and are recognizable across all domains of life.
- **Figure 1-17:** Shows the conservation of the rRNA gene sequence across an archaeon, a bacterium, and a human.

New Genes Are Generated from Preexisting Genes

- **Mechanisms of Genetic Innovation:**
  1. **Intragenic Mutation:** Modification of an existing gene.
  2. **Gene Duplication:** Accidental duplication of a gene, allowing the copies to diverge and specialize.
  3. **DNA Segment Shuffling:** Recombination of segments from different genes to create a hybrid gene.
  4. **Horizontal (Intercellular) DNA Transfer:** Transfer of DNA from one cell to another, even between species.
- **Figure 1-18:** Illustrates these four modes of genetic innovation.

Gene Duplications Give Rise to Families of Related Genes

- **Gene Families:** A set of related genes within a single genome that arose from one ancestral gene through duplication and divergence.
- **Homologs:** Genes related by descent.
  - **Orthologs:** Genes in different species that evolved from a common ancestral gene (e.g., human hemoglobin and chimpanzee hemoglobin). They typically retain the same function.
  - **Paralogs:** Genes within a single genome that resulted from a duplication event (e.g., different hemoglobin genes in humans). They often have diverged in function.
- **Figure 1-19:** Shows the prevalence of gene families in the **Bacillus subtilis** genome.
- **Figure 1-20:** Distinguishes between orthologs and paralogs.

The Function of a Gene Can Often Be Deduced from Its Nucleotide Sequence

- By comparing the sequence of a new gene to databases of known genes, one can find homologs whose functions have been determined experimentally. This allows for an educated guess about the new gene’s function.

More Than 200 Gene Families Are Common to All Three Domains of Life

- Systematic searches for homologous genes across all three domains have identified hundreds of ancient conserved gene families, likely present in the last universal common ancestor.
- **Table 1-1:** Lists the number of universal gene families, with the largest numbers involved in translation and amino acid metabolism.

Eukaryotes and the Origin of the Eukaryotic Cell

Eukaryotic cells are larger and more complex than prokaryotic cells, containing a nucleus and other organelles.

Eukaryotic Cells Contain a Variety of Organelles

- **Nucleus:** Contains the cell’s DNA, enclosed by a double-membrane nuclear envelope perforated by nuclear pores.
- **Cytoskeleton:** A network of protein filaments (actin filaments, microtubules, intermediate filaments) that provides mechanical strength, controls cell shape, and drives cell movements.
- **Membrane-Enclosed Organelles:**
  - **Endoplasmic Reticulum (ER):** Synthesizes most cell membrane components and proteins for export.
  - **Golgi Apparatus:** Modifies and packages molecules from the ER.
  - **Lysosomes:** Site of intracellular digestion.
  - **Peroxisomes:** Use hydrogen peroxide to inactivate toxic molecules.
  - **Mitochondria:** Generate most of the cell’s ATP through cellular respiration.
  - **Chloroplasts:** (in plants and algae) Perform photosynthesis.
- **Cytosol:** The fluid surrounding the organelles, where many metabolic reactions and protein synthesis (on ribosomes) occur.
- **Endocytosis and Exocytosis:** Processes for importing and exporting materials via vesicles.
- **Figure 1-21:** A diagram showing the major organelles of a typical animal cell.
- **Figure 1-22 & 1-23:** Illustrate phagocytosis, endocytosis, and exocytosis.

Mitochondria and Chloroplasts Evolved from Symbiotic Bacteria

- **Endosymbiotic Theory:** Proposes that mitochondria and chloroplasts originated as free-living bacteria that were engulfed by an ancestral host cell.
- **Evidence for Endosymbiosis:**
  - Both organelles are similar in size to bacteria.
  - They contain their own circular DNA genomes, ribosomes, and translation machinery, all of which resemble those of bacteria.
  - They reproduce by dividing, similar to bacteria.
- **Origin of Mitochondria:** An ancestral anaerobic archaeon is thought to have captured an aerobic bacterium, establishing a symbiotic relationship. The host provided protection and nutrients, while the bacterium provided efficient ATP production.
  - **Asgard Archaea:** A recently discovered lineage of archaea with many eukaryotic-like genes and complex cell protrusions, providing a plausible model for the ancestral host.
- **Origin of Chloroplasts:** An early eukaryotic cell that already had mitochondria later engulfed a photosynthetic bacterium.

- **Figure 1-25 & 1-28:** Show the structure of a mitochondrion and a chloroplast.
- **Figure 1-26 & 1-27:** Show an Asgard archaeon and a model for its role in eukaryotic evolution.
- **Figure 1-29:** A model for the evolution of eukaryotic cells, showing the sequential acquisition of mitochondria and then chloroplasts.

Eukaryotes Have Hybrid Genomes

- The eukaryotic nuclear genome is a hybrid, containing genes from both its ancestral archaeon host and its bacterial endosymbionts.
- Many genes originally from the mitochondrial and chloroplast genomes have been transferred to the host cell's nucleus over evolutionary time.

Eukaryotic Genomes Are Big and Rich in Regulatory DNA

- **Genome Size:** Eukaryotic genomes are typically hundreds of times larger than prokaryotic ones.
- **Noncoding DNA:** A large fraction of eukaryotic DNA does not code for protein or RNA (e.g., 98.5% in humans vs. 11% in **E. coli**).
- **Regulatory DNA:** Much of this noncoding DNA is involved in the complex regulation of gene expression, which is crucial for the development and function of multicellular organisms.
- **Figure 1-30 & Table 1-2:** Compare genome sizes and gene numbers across different organisms.

Eukaryotic Genomes Define the Program of Multicellular Development

- **Cell Differentiation:** In a multicellular organism, all cells contain the same genome but express different sets of genes, leading to specialized cell types (e.g., nerve cells, skin cells).
- **Developmental Program:** The genome contains the "software" (regulatory DNA and genes for regulatory proteins/RNAs) that controls how the cellular "hardware" is used, directing the complex processes of embryonic development.
- **Figure 1-31 & 1-32:** Show the diversity of cell types and an example of how a mutation in a regulatory gene affects development.

Many Eukaryotes Live as Solitary Cells

- Single-celled eukaryotes (protists) are incredibly diverse and can be as complex in their structure and behavior as multicellular organisms.
- They include hunters (e.g., **Didinium**), photosynthesizers, and scavengers.
- Genome comparisons show that single-celled eukaryotes are more evolutionarily diverse than plants and animals, which arose relatively late.
- **Figure 1-33 & 1-34:** Display the variety and complex behavior of single-celled eukaryotes.
- **Figure 1-35:** A eukaryotic tree of life showing the vast diversity of single-celled lineages.

Model Organisms

Biologists focus on a few model organisms to understand universal biological principles.

Molecular Biology Began with a Spotlight on E. coli and Its Viruses

- **Escherichia coli (E. coli):** A bacterium that has been a cornerstone of molecular biology. It is easy to grow, reproduces rapidly (every 20 mins), and has a relatively simple genome (4.6 million nucleotide pairs, 4300 genes).
- **Viruses:** Parasitic genetic elements that depend on host cells for reproduction.
  - **Bacteriophages:** Viruses that infect bacteria. Studies of bacteriophages T4 and lambda were critical for understanding DNA replication, gene regulation, and biological assembly.
- **Horizontal Gene Transfer:** Viruses and the natural ability of bacteria to take up DNA from the environment facilitate the transfer of genes between cells, a major driver of bacterial evolution.
- **Figure 1-37 & 1-38:** Show the T4 bacteriophage and the genome of **E. coli**.

A Yeast Serves as a Minimal Model Eukaryote

- **Saccharomyces cerevisiae (Budding Yeast):** A simple, single-celled fungus used as a model for eukaryotic cell biology.
- **Features:** Small genome (12.5 million nucleotide pairs, 6600 genes), rapid growth, and amenability to genetic manipulation. It can reproduce both asexually and sexually.
- **Contributions:** Crucial for understanding the cell-division cycle, meiosis, chromosome structure, and many other fundamental eukaryotic processes.
- **Figure 1-39 & 1-40:** Show the yeast cell and its reproductive cycles.

Arabidopsis thaliana Has Been Chosen as a Model Plant

- A small weed in the cabbage family, chosen as the primary model for plant biology.
- **Features:** Small genome for a plant (135 million nucleotide pairs), rapid life cycle (8-10 weeks), and easy to grow indoors.
- **Contributions:** Understanding flower development, plant hormones, and plant immunity.
- **Figure 1-41:** Shows the **Arabidopsis** plant.

The World of Animal Cells Is Represented by a Worm, a Fly, a Fish, a Mouse, and a Human

- **Caenorhabditis elegans (Nematode Worm):** A simple animal with a fixed number of body cells (959). Its transparent body and predictable development make it ideal for studying cell lineage and processes like programmed cell death (apoptosis).
  - **Figure 1-42:** Shows the **C. elegans** worm.
- **Drosophila melanogaster (Fruit Fly):** A classic model for genetics. Studies in **Drosophila** provided foundational proof that genes are on chromosomes and have been key to understanding animal development.
  - **Figure 1-43 & 1-44:** Show a normal and mutant fly, and its giant polytene chromosomes.
- **Danio rerio (Zebrafish):** A vertebrate model with a transparent embryo that develops outside the mother, allowing for direct observation of development, especially of the circulatory system.
  - **Figure 1-46:** Shows a zebrafish and its developing embryo.
- **Mus musculus (Mouse):** The foremost mammalian model organism. Its genetics and physiology are very similar to humans, and powerful techniques exist to create targeted mutations to study gene function in the context of a whole animal.
  - **Figure 1-48:** Shows how a similar mutation in the **Kit** gene causes the same white-forelock phenotype in both mouse and human.

- **Homo sapiens (Human):** Humans serve as a unique model due to the vast amount of self-reported medical data and the large, diverse population, which provides a rich database of mutations and their effects.

The COVID-19 Pandemic Has Focused Scientists on the SARS-CoV-2 Coronavirus

- **SARS-CoV-2:** A coronavirus with a large single-strand RNA genome (30,000 nucleotides, 29 proteins).
- **Structure:** The RNA genome is packaged in a protein coat and a lipid envelope with protruding spike proteins.
- **Function:** The spike protein attaches to host cells, enabling viral entry. Other viral proteins replicate the genome, package new virus particles, and help evade the host's immune system.
- **Figure 1-49:** Shows the structure and protein-coding genes of SARS-CoV-2.